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EXPERIMENT 17

Edge Detection using the Sobel Operator along the X Axis

AIM

To perform edge detection on the given image using the Sobel operator along the X axis.

PROGRAM

```
import cv2

img = cv2.imread("image1.jpg", cv2.IMREAD_GRAYSCALE)

# CV_64F keeps negative gradients, which are lost if CV_8U is used directly
sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=5)
sobel_x = cv2.convertScaleAbs(sobel_x)      # back to 8-bit for display

cv2.imwrite("sobel_x.jpg", sobel_x)
cv2.imshow("Original", img)
cv2.imshow("Sobel X", sobel_x)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT



image1.jpg (input image)

OUTPUT



sobel_x.jpg (vertical edges detected by the X gradient)

RESULT

Thus the edges along the X axis were successfully detected using the Sobel operator.